



Split-Flap Display



David Kingsman

[VIEW IN BROWSER](#)

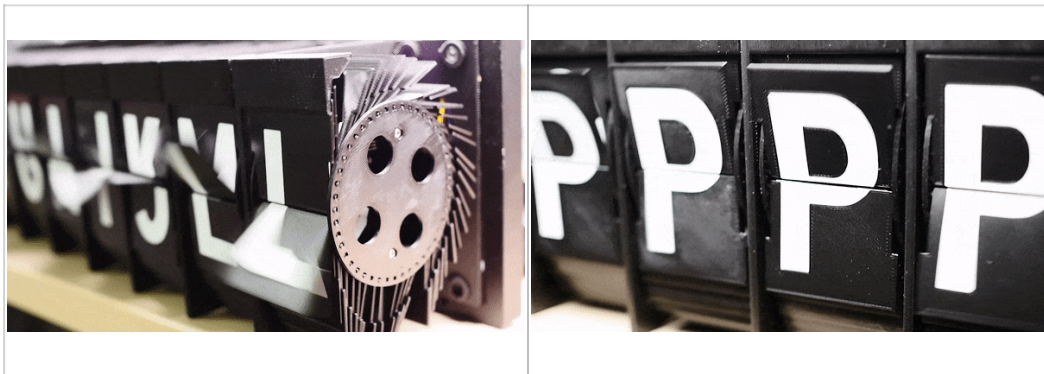
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Summary

10-unit split-flap display. 3D-printed flaps. Modular.

[Hobby & Makers](#) > [Electronics](#)

Tags: [display](#) [splitflap](#) [arduino](#) [esp8266](#)



This split-flap display sets itself apart by being almost completely 3D-printed. The flaps are printed by changing the color mid-print and mid-layer. The case is also completely 3D-printed.

Video of the display in action (with sound):

https://www.reddit.com/r/3Dprinting/comments/o4boxk/i_designed_my_own_splitflap_display/

It can be used as a date display, clock or anything really. It is connected to the local network with an esp8266.

Josh Murrah made an in-depth video about his build:

General info about display:

- 10 units (pcb design allows up to 16)
- 45 flaps per unit
- Size: 513 mm x 136 mm x 138 mm

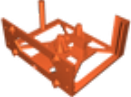
More info in the instruction manual (scroll all the way down to “other files” in the files tab right here on prusaprinters).

GitHub: <https://github.com/Dave19171/split-flap>

Partially inspired by <https://splitflap.home.blog/>

Model files

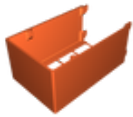
 **Unit** 3 files

**unitframe.stl**

**flapdruminner.stl**

**flapdrumouter.stl**

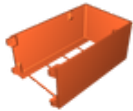
 **FrontCover** 3 files



frontcoverleft.stl



frontcovercenter.stl

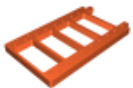


frontcoverright.stl



MiddleFrame

3 files



middleframeleft.stl



middleframecenter.stl

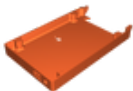


middleframeright.stl

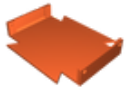


BackCover

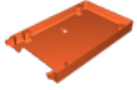
3 files



backcoverleft.stl



backcovercenter.stl



backcoverright.stl

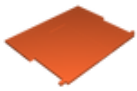


Flaps 1-9

26 files



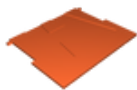
flap1-9.3mf



flap1.stl



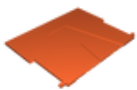
flap1_1.stl



flap2.stl



flap2_1.stl



flap3.stl



flap3_1.stl



flap32.stl



flap4.stl



flap4_1.stl



flap4_2.stl



flap5.stl



flap5_1.stl



flap5_2.stl



flap6.stl



flap6_1.stl



flap6_2.stl



flap7.stl



flap7_1.stl



flap7_2.stl



flap8.stl



flap8_1.stl



flap8_2.stl



flap9.stl



flap9_1.stl



flap9_2.stl



Flaps 10-18

33 files



flap10-18.3mf



flap10.stl



flap10_1.stl



flap10_2.stl



flap11.stl



flap11_1.stl



flap11_2.stl



flap12.stl

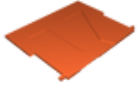


flap12_1.stl

flap12_2.stl



flap13.stl



flap13_1.stl



flap13_2.stl



flap14.stl



flap14_1.stl



flap14_2.stl



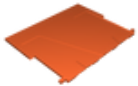
flap14_3.stl



flap14_4.stl



flap15.stl



flap15_1.stl



flap15_2.stl



flap15_3.stl



flap15_4.stl



flap16.stl



flap16_1.stl



flap16_2.stl



flap16_3.stl



flap17.stl



flap17_1.stl





flap17_2.stl



flap18.stl



flap18_1.stl



flap18_2.stl



Flaps 19-27

33 files



flap19-27.3mf



flap19.stl



flap19_2.stl



flap19_1.stl



flap20.stl



flap20_1.stl



flap20_2.stl



flap22_1.stl



flap21.stl



flap21_1.stl



flap21_2.stl



flap22.stl



flap22_2.stl



flap23.stl

flap23_1.stl



flap23_2.stl



flap23_3.stl



flap24.stl



flap24_1.stl



flap24_2.stl



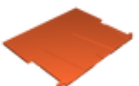
flap24_3.stl



flap24_4.stl



flap25.stl



flap25_1.stl



flap25_2.stl



flap25_3.stl



flap25_4.stl



flap26.stl



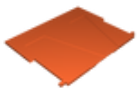
flap26_1.stl



flap26_2.stl



flap27.stl



flap27_1.stl



flap27_2.stl





Flap 28-36 no umlaute

13 files



flap28-36_no_umlaut.3mf



flap28_no_um.stl



flap28_1_no_um.stl



flap28_2_no_um.stl



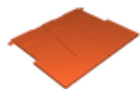
flap29_no_um.stl



flap29_1_no_um.stl



flap29_2_no_um.stl



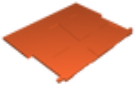
flap30_no_um.stl



flap30_1_no_um.stl



flap30_2_no_um.stl



flap31_no_um.stl



flap31_1_no_um.stl



flap31_2_no_um.stl

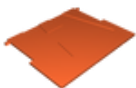


Flaps 28-36

35 files



flap28-36.3mf



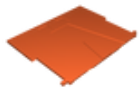
flap28.stl



flap28_1.stl



flap28_2.stl



flap29.stl



flap29_1.stl



flap29_2.stl



flap29_3.stl



flap29_4.stl



flap30.stl



flap30_1.stl



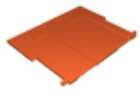
flap30_2.stl



flap30_3.stl



flap30_4.stl



flap31.stl



flap31_1.stl



flap31_2.stl



flap31_3.stl



flap31_4.stl



flap31_5.stl



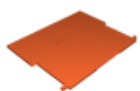
flap32.stl



flap32_1.stl



flap32_2.stl



flap33.stl



flap33_1.stl



flap33_2.stl



flap34.stl



flap34_1.stl



flap34_2.stl



flap35.stl



flap35_1.stl



flap35_2.stl



flap36.stl



flap36_1.stl



flap36_2.stl



Flaps 37-45

28 files



flap37-45.3mf



flap37.stl



flap37_1.stl



flap37_2.stl



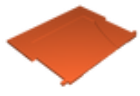
flap38.stl



flap38_1.stl



flap38_2.stl



flap39.stl



flap39_1.stl



flap39_2.stl



flap40.stl



flap40_1.stl



flap40_2.stl



flap41.stl



flap41_1.stl



flap41_2.stl



flap42.stl

flap42_1.stl



flap42_2.stl



flap43.stl



flap43_1.stl



flap44.stl



flap44_1.stl



flap44_2.stl



flap45.stl



flap45_1.stl

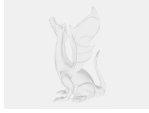


flap45_2.stl





flap45_3.stl

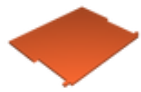


assembly.step



flap_blank.step

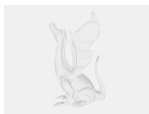
☐ If you want to make your own custom flaps



flap_blank_do_not_print.stl

☐ If you want to make your own custom flaps

Other files



splitflapinstructions.pdf

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